

Ontological Collapse: What the Classical Atom Reveals About Scientific Models and the Metaphysics They Carry

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Abstract

We present a discussion on the classical calculation of the radiative collapse of the hydrogen atom, and how it constitutes not merely an empirical failure but an *ontological* one. The planetary model of the atom fails not because its equations are internally inconsistent, but because the type of entities it presupposes—such as a point charge with a definite trajectory—does not correspond to what electrons are. We use this case to distinguish two layers present in any physical model, often ignored or not sufficiently emphasize: its predictive mathematical content and its implicit ontological commitments. We then argue that Aristotle’s distinction between potentiality (δύναμις) and actuality (ἐνέργεια), recovered by Heisenberg in 1958, provides a more adequate ontological framework for the quantum atom than either naive particle realism or strict instrumentalism. The paper closes by situating this reading within contemporary debates on structural realism and the metaphysics of quantum objects.

1 The Classical Collapse: An Explicit Calculation

The failure of the planetary models of the atom is a well known and extensively discussed example. We present the calculation in full because the philosophical argument of Chapter [...] depends on it: the failure is not a rough estimate but an exact consequence of three internally consistent and independently verified physical laws.

1.1 The Three Ingredients and Their Sources

Centripetal acceleration (Newton, 1687)

Newton's *Principia Mathematica* [1] establishes that for uniform circular motion of radius r and speed v , differentiating the position vector $\vec{r}(t) = r(\cos \omega t, \sin \omega t)$ twice with $\omega = v/r$ gives

$$\ddot{\vec{r}} = -\omega^2 \vec{r} \implies \boxed{a = \frac{v^2}{r}}, \quad (1)$$

directed toward the centre. No approximation is involved.

Coulomb's law (Coulomb, 1785)

Measured by torsion balance in 1785 [2], the force arising between two point charges q_1 and q_2 separated by distance r is

$$\boxed{F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}}, \quad (2)$$

attractive for opposite charges. The constant ϵ_0 would be added by Maxwell's 1865; the inverse-square dependence is Coulomb's direct experimental result.

Larmor radiation formula (Larmor, 1897)

From Maxwell's equations, the electric field radiated by an accelerated charge falls as $1/r$, so the Poynting flux integrated over a distant sphere yields a finite radiated power [3]:

$$\boxed{P = \frac{e^2 a^2}{6\pi\epsilon_0 c^3}}. \quad (3)$$

The dependence on a^2 follows because the radiated field is linear in a , and power is quadratic in the field amplitude.

1.2 The Differential Equation for $r(t)$

Setting Coulomb's force equal to the centripetal requirement for the electron (mass m_e , charge e) one obtains:

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r} \implies v^2 = \frac{e^2}{4\pi\epsilon_0 m_e r}. \quad (4)$$

The total mechanical energy is then:

$$E = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}, \quad (5)$$

where the virial theorem ensures $E_{\text{kin}} = -E/2$.

As the electron loses energy by radiation, E decreases and r decreases monotonically. Substituting $a = e^2/(4\pi\epsilon_0 m_e r^2)$ into (3):

$$P = \frac{e^6}{12\pi^3 \epsilon_0^3 m_e^2 c^3 r^4}. \quad (6)$$

Equating total energy $\dot{E} = -P$ and differentiating (5) one obtains:

$$\frac{e^2}{8\pi\epsilon_0 r^2} \dot{r} = -\frac{e^6}{12\pi^3\epsilon_0^3 m_e^2 c^3 r^4} . \quad (7)$$

Rearranging:

$$r^2 dr = -\frac{2e^4}{3(4\pi\epsilon_0)^2 m_e^2 c^3} dt \equiv -C dt . \quad (8)$$

1.3 Integration and Result

Integrating (8) from $r_0 = a_0$ (the Bohr radius, $a_0 \approx 0.529 \text{ \AA}$) to $r = 0$:

$$\int_{a_0}^0 r^2 dr = -C t_{\text{collapse}} \implies \frac{a_0^3}{3} = C t_{\text{collapse}}, \quad (9)$$

giving the final results

$$t_{\text{collapse}} = \frac{(4\pi\epsilon_0)^2 m_e^2 c^3 a_0^3}{4 e^4} \approx 1.6 \times 10^{-11} \text{ s} . \quad (10)$$

Sixteen picoseconds. This is not an order-of-magnitude estimate; it is the exact result of integrating three independently verified and internally consistent physical laws. The calculation is impeccable, yet the conclusion is empirically false. That tension is the starting point of everything that follows.

Constant	Symbol	Value
Bohr radius	a_0	$0.529 \times 10^{-10} \text{ m}$
Electron mass	m_e	$9.109 \times 10^{-31} \text{ kg}$
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Speed of light	c	$2.998 \times 10^8 \text{ m s}^{-1}$
Permittivity of vacuum	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$

Table 1: Physical constants used in the collapse calculation.

2 Two Layers in Every Physical Model

The collapse calculation exemplifies a general structure present in all physical models, but rarely made explicit. Every model carries at least two distinguishable layers of content:

1. **A predictive-mathematical layer:** the equations, boundary conditions, and inference rules that generate quantitative predictions about measurable quantities.
2. **An ontological layer:** the implicit commitments about what kinds of entities exist, what properties they possess independently of measurement, and how they relate causally.

Standard empiricist philosophy of science focuses almost exclusively on the first layer [4]. A model is adequate if its predictions agree with experiment; inadequate otherwise. On this view, the classical atomic model fails because it predicts a collapse timescale of $\sim 10^{-11}$ s while atoms are stable indefinitely. That diagnosis is correct, but incomplete.

2.1 The Ontological Layer of the Classical Model

The deeper diagnosis is that the classical model fails at the second layer. Newton’s equations for dynamics, Coulomb’s law for electromagnetism, and Larmor’s radiation formula are not wrong in themselves—they remain excellent approximations for macroscopic charged objects. Yet, they fail for the electron in the hydrogen atom because they embed a specific ontology:

- Electrons are *point particles* with definite position and momentum at every instant.
- They move along *continuous trajectories* governed by deterministic force laws.
- They act as *classical sources* for the electromagnetic field, coupling through their charge and acceleration.

It is known nowadays that each of these assumptions is false. A model can therefore be *mathematically consistent*, *empirically adequate in its domain*, and *ontologically erroneous*. The classical hydrogen atom is a case study in exactly this combination.

2.2 The General Diagnostic Distinction

We propose the following general principle:

When a physical model fails empirically at the boundary of its domain, one must ask whether the failure is predictive or ontological. A predictive failure can often be corrected by modifying equations within the same ontological framework. An ontological failure requires replacing the picture of what exists.

As an example, the transition from Newtonian to relativistic mechanics is largely *predictive*: the ontology of point particles with definite trajectories is preserved; only force laws and the metric of spacetime change. The transition from classical to quantum mechanics is *ontological*: the very category of “particle with a definite trajectory” ceases to apply.

This distinction has implications for how we interpret theoretical progress. Kuhnian paradigm shifts [5] and Lakatosian research programmes [6] both capture aspects of it, but neither systematically separates the predictive from the ontological layer. The collapse calculation makes the separation vivid: the predictive machinery is flawless; it is the ontological presuppositions that must go.

[TODO: Expand: Cartwright on laws vs. capacities; Giere on model-world relations; Chakravartty on semi-realism. Key point: none of these frameworks isolates the ontological layer as a separate diagnostic target in quite the way we propose.]

3 Bohr’s Provisional Resolution and the Persistence of the Problem

3.1 The 1913 Postulates as Ontological Stipulation

Bohr’s response in 1913 was philosophically unusual in its candour [7]. He did not attempt to repair the classical ontology. He declared classical electrodynamics inapplicable to intra-atomic orbits and introduced quantised stationary states as irreducible postulates: the electron in a stationary state does not radiate, not because some mechanism suppresses radiation, but because the very concept of radiating simply does not apply to that state.

This is an ontological statement, not a dynamical explanation. Bohr acknowledged as much, and indeed he described his postulates as provisional, awaiting a deeper theory. The immediate success was spectacular. The energy levels

$$E_n = -\frac{m_e e^4}{2\hbar^2} \cdot \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots \quad (11)$$

reproduce the Balmer series [8] and the full Rydberg formula [9], with the Rydberg constant now derived rather than fitted:

$$R_H = \frac{m_e e^4}{4\pi\hbar^3 c} \approx 1.097 \times 10^7 \text{ m}^{-1}. \quad (12)$$

The precision was stunning. Yet, the ontological justification was absent.

3.2 Quantum Mechanics: Relocation, Not Dissolution

Full quantum mechanics relocates rather than dissolves the problem. In the Schrödinger picture, the ground-state electron occupies the state $\psi_{1s}(r) \propto e^{-r/a_0}$, spherically symmetric. There is no definite orbit, no classical centripetal acceleration, and therefore no Larmor radiation. Stability is explained: the ground state is the lowest eigenvalue of the Hamiltonian $\hat{H} = \hat{p}^2/2m_e - e^2/4\pi\epsilon_0 r$, and there is no lower state to decay into.

Quantum electrodynamics (QED) goes further, correctly computing excited-state lifetimes, the Lamb shift (~ 1057 MHz for the $2s-2p_{1/2}$ splitting, measured by Lamb and Retherford in 1947 [10]), and spontaneous emission rates at parts-per-billion precision.

3.3 The Tail That Remains Open

Yet the problem has a residue. The self-energy of the electron in its own electromagnetic field has no fully satisfactory formulation. The Abraham–Lorentz equation [11, 12, 13],

$$m_e \dot{v} = F_{\text{ext}} + \frac{e^2}{6\pi\epsilon_0 c^3} \ddot{v}, \quad (13)$$

admits *runaway solutions* (acceleration growing exponentially without external force) and *pre-acceleration solutions* (particle responds before the force arrives, violating causality). In QED these pathologies are controlled by renormalisation, which works numerically with extraordinary precision but which Dirac regarded as a fundamental logical inconsistency [13] and Feynman called sweeping the problem under the rug with an elegant technique [14].

The ontological question Bohr implicitly raised in 1913—what *kind of thing* is an electron, considered as both a quantum object and a source of its own field?—remains incompletely answered. Each theoretical layer has provided a more adequate partial ontology. None has provided a complete one.

Consider a paragraph on effective field theory (Weinberg, Wilson): EFT explicitly embraces ontological incompleteness at each scale. That may be a more honest framing than “the problem is unsolved.”

4 Aristotle’s Potentiality and the Quantum State

4.1 Heisenberg’s Invocation

In *Physics and Philosophy* (1958), Heisenberg introduced an Aristotelian category to interpret quantum mechanics [15]:

“[The probability wave] was a quantitative version of the old concept of *potentia* in Aristotelian philosophy. It introduced something standing in the middle between the idea of an event and the actual event, a strange kind of physical reality just in the middle between possibility and reality.”

This invocation is often treated as a philosophical metaphor appended to an otherwise self-contained physical theory. We argue it is more: it is a serious ontological proposal that deserves reconstruction and critical evaluation on its own terms.

4.2 The Aristotelian Framework

Aristotle’s *Metaphysics* [16] distinguishes two fundamental modes of being:

- δύναμις (potentiality): what a thing can be or do, considered as a real but not yet actualised property. Not mere logical possibility, nor ignorance about an already determinate state, but a genuine ontological category—a real property of things, prior to and distinct from their actual states.
- ἐνέργεια / ἐντελέχεια (actuality): the realisation of that potentiality; what a thing is in full exercise of its nature.

For Aristotle, bronze *is* potentially a statue before any sculptor acts; the seed *is* potentially a tree. These are not descriptions of what we happen not to know; they describe what the thing really is, in a mode of being distinct from full actuality. Such categorization of nature and its entities [ontos on ..., Zubiri y la mirada ciega de quien ignora la etimología ...]

4.3 Application to the Quantum State

The mapping is structurally precise. Before a position measurement on an electron in state ψ , the electron does not have a definite position we merely fail to know—this is what Bell’s theorem and its experimental confirmation rule out [17, 18, 19].

Rather, the electron has a distribution of *potential* positions, encoded by the so-called wave function $|\psi(x)|^2$, which are real properties of the electron’s present state. Measurement does not reveal a pre-existing value; it *actualises* one of the potentialities.

On this reading:

- ψ is the mathematical representation of the electron’s potentialities.
- Measurement is the physical process of passage from potentiality to actuality.
- The Born rule $|\psi(x)|^2 dx$ gives the quantitative structure of the potentiality distribution.

This is not the Copenhagen interpretation in its instrumentalist form, which treats ψ as a bookkeeping device for predictions. It is a realist reading: the potentialities are real features of the world, not features of our knowledge.

4.4 What This Resolves and What It Does Not

The Aristotelian reading dissolves the puzzle of the classical collapse. The $1s$ electron has no definite trajectory; there is no classical kinematic state to insert into Larmor’s formula (3). The question “why does the orbiting electron not radiate?” does not arise, because its presupposition—that the electron has a definite centripetal acceleration—is false.

What this reading does *not* resolve is the measurement problem in its general form. It relocates the hard question to the actualisation process: what determines which potentiality becomes actual? The major interpretations map partially onto different answers:

- **Copenhagen:** actualisation is a primitive, theory-external fact; quantum mechanics describes potentialities but not the process of actualisation itself.
- **GRW:** spontaneous collapse provides a physical mechanism for actualisation, at the cost of modifying the Schrödinger equation [20].
- **Everett:** all potentialities are actualised in branching worlds; no collapse occurs [21].
- **Bohmian mechanics:** particles always have definite positions (full actuality); the pilot wave guides them deterministically [22].

We do not adjudicate between these here. Our claim is more modest: the Aristotelian framework provides a more adequate *starting* ontology for quantum mechanics than either classical particle realism or strict instrumentalism, and the collapse calculation is the sharpest way to see why.

[TODO: Reviewer will ask: is potentiality realism distinguishable from an epistemic interpretation? Key argument: potentialities are ontic, not epistemic—they ground the Born-rule probabilities rather than describing ignorance. See [23] and [24].]

5 Entanglement, Structural Realism, and Conclusion

5.1 Entanglement as Irreducibly Relational Potentiality

The potentiality reading connects naturally to ontic structural realism (OSR) [25, 26]. OSR proposes that what is real are structures—patterns of relations—rather than objects with intrinsic natures, motivated in part by the permutation symmetry of identical particles [27] and by the non-locality revealed by Bell inequality violations [17, 18, 19].

Consider the spin-singlet state of two electrons,

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B). \quad (14)$$

This state cannot be written as a product $|\phi\rangle_A \otimes |\chi\rangle_B$. Neither particle A nor B has a definite spin potentiality on its own. On the Aristotelian reading, the potentialities are properties not of A and B separately but of the *pair* AB jointly—an irreducibly relational structure.

Bell’s theorem [17] shows that no local hidden-variable theory can reproduce the quantum correlations. The CHSH inequality

$$|E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2 \quad (15)$$

is violated by quantum mechanics up to $2\sqrt{2}$ (the Tsirelson bound), confirmed experimentally [18, 19]. Entanglement is not a correlation between independently real objects. It is a structure of irreducibly joint potentiality with no local decomposition—precisely the structural realist’s claim that relations are ontologically prior to relata.

5.2 Potentiality Realism vs. Ontic Structural Realism

The question arises whether the Aristotelian potentiality reading simply *is* a form of OSR. We suggest it occupies a distinct position. OSR in its strongest form eliminates objects entirely in favour of structures [27]. The Aristotelian reading preserves objects—the electron is a genuine entity—but denies that objects are fully actual with respect to all their properties prior to measurement.

This is a form of *potentiality realism*: realism about both objects and their potentialities, with actuality restricted to what has been, or is being, measured. Objects are real; their potentialities are real; their actualities are contingent on measurement interactions.

[TODO: Engage with Esfeld and Deckert’s minimalist ontology [28]: points in a structure with no intrinsic properties. Argue potentiality realism is distinct: it accepts intrinsic properties *in potentia*, not *in actu*.]

5.3 Conclusion

The classical calculation of the hydrogen atom’s radiative collapse is more than a historical curiosity. It is a precise, mathematically impeccable derivation—from Newton, Coulomb, and Larmor, all independently confirmed in their domains—that produces an empirically false result. And the reason it is false is not a flaw in the equations. It is a flaw in the ontology they presuppose.

We have argued for three connected claims. First, every physical model carries an ontological layer analytically distinct from its predictive-mathematical content; diagnosing which layer has failed is a non-trivial philosophical task that physics alone cannot perform. Second, Bohr’s 1913 resolution was an explicit ontological stipulation—he declared the classical ontology

inapplicable without fully replacing it—and the residue of this incompleteness persists in the renormalisation programme of QED. Third, Aristotle’s potentiality/actuality distinction, recovered by Heisenberg in his 1958 essay, provides a more adequate starting ontology for quantum mechanics than either classical particle realism or strict instrumentalism. Finally we discuss entanglement, which read through this lens, is a structure of irreducibly joint potentiality that connects naturally to ontic structural realism while preserving a genuine role for objects.

The lesson is general. Scientific progress is not only the accumulation of more accurate predictions. It is the successive replacement of ontologies—each adequate to a domain, each carrying implicit metaphysical commitments, each eventually confronted with phenomena that force not just new equations but a new picture of what there is.

Sixteen picoseconds. That is how long a classical hydrogen atom would survive. That it does not survive that way, and that we needed a century of physics and philosophy to understand why, says something important about the relationship between mathematics, experiment, and the picture of the world that any physical theory always—whether acknowledged or not—carries with it.

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